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Help

The **due date** for this quiz is **Mon 27 Jan 2014 7:59 PM PST**.

- [ ] **In accordance with the Coursera Honor Code, I (Sandipan Dey) certify that the answers here are my own work.**

Question 1

Overfitting and Deterministic Noise

Deterministic noise depends on $\mathcal{H}$, as some models approximate f better than others. Assume $\mathcal{H}' \subset \mathcal{H}$ and that f is fixed. \textbf{In general} (but not necessarily in all cases), if we use $\mathcal{H}'$ instead of $\mathcal{H}$, how does deterministic noise behave?

- In general, deterministic noise will be the same.
- There is no trend.
- In general, deterministic noise will decrease.
- In general, deterministic noise will increase.

Question 2

Regularization With Polynomials. Polynomial models can be viewed as linear models in a space $\mathcal{Z}$, under a nonlinear transform $\Phi : \mathcal{X} \to \mathcal{Z}$. Here, Φ transforms the scalar x into a vector $\mathbf{z}$ of Legendre polynomials, $\mathbf{z} = (1, L_1(x), L_2(x), \ldots, L_Q(x))$. Our hypothesis set will be expressed as a linear combination of these polynomials,

$\mathcal{H}_Q = \left\{ h \mid h(x) = \mathbf{w}^{\mathrm{T}}\mathbf{z} = \sum_{q=0}^{Q} w_q L_q(x) \right\}$, where $L_0(x) = 1$.

Consider the following hypothesis set defined by the constraint:

$\mathcal{H}(Q, c, Q_o) = \{ h \mid h(x) = \mathbf{w}^{\mathrm{T}}\mathbf{z} \in \mathcal{H}_Q; w_q = c \text{ for } q \geq Q_o \}$, which of the following statements is correct?

- $\mathcal{H}(10,1,3) \cap \mathcal{H}(10,1,4) = \mathcal{H}_1$
- $\mathcal{H}(10,0,3) \cap \mathcal{H}(10,0,4) = \mathcal{H}_2$
- $\mathcal{H}(10,0,3) \cup \mathcal{H}(10,1,4) = \mathcal{H}_3$

- $\mathcal{H}(10,0,3) \cup \mathcal{H}(10,0,4) = \mathcal{H}_4$

Question 3

Regularization and Weight Decay. Consider the augmented error

$$E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\,\mathbf{w}^T\,\mathbf{w}$$

with some $\lambda > 0$.

If we want to minimize the augmented error $E_{\text{aug}}(\mathbf{w})$ by gradient descent, with η as learning rate, which of the followings are the correct update rules?

- [ ] $\mathbf{w}(t+1) \longleftarrow (1-\frac{\eta\lambda}{N})\mathbf{w}(t) - \eta\nabla E_{\text{in}}(\mathbf{w})$.
- [ ] $\mathbf{w}(t+1) \longleftarrow (1-\frac{2\eta\lambda}{N})\mathbf{w}(t) - \eta\nabla E_{\text{in}}(\mathbf{w})$.
- [ ] $\mathbf{w}(t+1) \longleftarrow \mathbf{w}(t) - \eta\nabla E_{\text{in}}(\mathbf{w})$.
- [ ] $\mathbf{w}(t+1) \longleftarrow \mathbf{w}(t) - \eta\nabla E_{\text{aug}}(\mathbf{w})$.

Question 4

Let $\mathbf{w}_{\text{lin}}$ be the optimal solution for the plain-vanilla linear regression and $\mathbf{w}_{\text{reg}}(\lambda)$ be the optimal solution for the formulation above. Select all the correct statements below:

- [ ] $\|\mathbf{w}_{\text{reg}}(\lambda)\| \geq \|\mathbf{w}_{\text{lin}}\|$ for any $\lambda > 0$
- [ ] $\|\mathbf{w}_{\text{reg}}(\lambda)\|$ is a non-decreasing function of λ for $\lambda \geq 0$
- [ ] $\|\mathbf{w}_{\text{reg}}(\lambda)\| \leq \|\mathbf{w}_{\text{lin}}\|$ for any $\lambda > 0$
- [ ] $\|\mathbf{w}_{\text{reg}}(\lambda)\|$ is a non-increasing function of λ for $\lambda \geq 0$

Question 5

Leave-One-Out Cross-Validation.

You are given the data points: $(-1,0), (\rho,1), (1,0),\ \rho \geq 0$, and a choice between two models: constant [$h_0(x) = b$] and linear [$h_1(x) = ax + b$]. For which value of ρ would the two models be tied using leave-one-out cross-validation with the squared error measure?

- $\sqrt{9-\sqrt{6}}$
- $\sqrt{\sqrt{3}-1}$
- $\sqrt{9+4\sqrt{6}}$
- $\sqrt{\sqrt{3}+4}$

Question 6

Learning Principles. Suppose that for 5 weeks in a row, a letter arrives in the mail that predicts the outcome of the upcoming Monday night baseball game.(Assume there are no tie.) You keenly watch each Monday and to your surprise, the prediction is correct each time. On the day after the fifth game, a letter arrives, stating that if you wish to see next week's prediction, a payment of NTD 1,000 is required.

Which of the following statements are true?

- [ ] There are 32 win-lose predictions for 5 games.
- [ ] If the sender wants to make sure that at least one person receives correct predictions on all 5 games from him, the sender should target to begin with at least 5 people.
- [ ] To make sure that at least one person receives correct predictions on all 5 games from him, after the first letter `predicts' the outcome of the first game, the sender should target 16 people with the second letter.
- [ ] To make sure that at least one person receives correct predictions on all 5 games from him, at least 64 letters are sent at the end of the 5 weeks.

Question 7

If the cost of printing and mailing out each letter is NTD 10. How much money can the sender make for the above 'fraud' to succeed once? That is, one of the recipients does send him NTD 1,000 to receive the prediction of the 6-th game?

- NTD 370
- NTD 400
- NTD 430
- NTD 340

Question 8

In our credit card example, the bank starts with some vague idea of what constitutes a good credit risk. So, as customers $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$ arrive, the bank applies its vague idea to approve credit cards for some of these customers based on a formula $a(\mathbf{x})$. Then, only those who get credit cards are monitored to see if they default or not. For simplicity, suppose that the first $N = 10,000$ customers were given credit cards by the credit approval function $a(\mathbf{x})$. Now that the bank knows the behavior of these customers, it comes to you to improve their algorithm for approving credit. The bank gives you the data $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$. Before you look at the data, you do mathematical derivations and come up with a credit approval function. You now test it on the data and, to your delight, obtain perfect prediction.

What is M, the size of your hypothesis set?

- 1
- 2^N
- We have no idea about it.
- N

Question 9

With such an M, what does the Hoeffding bound say about the probability that the true average error rate of g is worse than 1% for $N = 10,000$?

- ≤ 0.171
- ≤ 0.221
- ≤ 0.271
- None of the other choices.

Question 10

You assure the bank that you have a got a system g for approving credit cards for new customers, which is nearly error-free. Your confidence is given by your answer to the previous question. The bank is thrilled and uses your g to approve credit for new customers. To their dismay, more than half their credit cards are being defaulted on. Assume that the customers that were sent to the old credit approval function and the customers that were sent to your g are indeed i.i.d. from the same distribution, and the bank is lucky enough (so the `bad luck' in the previous problem does not happen). Select all the true claims for this situation.

- [ ] By applying $a(\mathbf{x})\ \mathrm{XOR}\ g(\mathbf{x})$ to approve credit for new customers, the performance of the overall credit approval system can be improved with guarantee provided by the previous problem.
- [ ] By applying $a(\mathbf{x})\ \mathrm{AND}\ g(\mathbf{x})$ to approve credit for new customers, the performance of the overall credit approval system can be improved with guarantee provided by the previous problem.
- [ ] If the old credit approval function was $a(\mathbf{x}) = +1$ (approve all customers), the situation should not happen.
- [ ] By applying $a(\mathbf{x})\ \mathrm{OR}\ g(\mathbf{x})$ to approve credit for new customers, the performance of the overall credit approval system can be improved with guarantee provided by the previous problem.

Question 11

Virtual Examples and Regularization

Consider linear regression with virtual examples. That is, we add K virtual examples $(\tilde{\mathbf{x}}_1, \tilde{y}_1), (\tilde{\mathbf{x}}_2, \tilde{y}_2), \ldots, (\tilde{\mathbf{x}}_K, \tilde{y}_K)$ to the training data set, and solve

$$\min_{\mathbf{w}} \frac{1}{N+K}\left(\sum_{n=1}^{N}(y_n - \mathbf{w}^T\mathbf{x}_n)^2 + \sum_{k=1}^{K}(\tilde{y}_k - \mathbf{w}^T\tilde{\mathbf{x}}_k)^2\right).$$

We will show that using some `special' virtual examples, which were claimed to be a possible way to combat overfitting in Lecture 9, is related to regularization, another possible way to combat overfitting discussed in Lecture 10. Let $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1 \tilde{\mathbf{x}}_2 \ldots \tilde{\mathbf{x}}_K]^T$, and $\tilde{\mathbf{y}} = [\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_K]^T$.

What is the optimal $\mathbf{w}$ to the optimization problem above, assuming that all the inversions exist?

- ( ) $(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T\mathbf{y} + \tilde{\mathbf{X}}^T\tilde{\mathbf{y}})$
- ( ) $(\mathbf{X}^T\mathbf{X} + \tilde{\mathbf{X}}^T\tilde{\mathbf{X}})^{-1}(\mathbf{X}^T\mathbf{y} + \tilde{\mathbf{X}}^T\tilde{\mathbf{y}})$
- ( ) $(\mathbf{X}^T\mathbf{X})^{-1}(\tilde{\mathbf{X}}^T\tilde{\mathbf{y}})$
- ( ) $(\mathbf{X}^T\mathbf{X} + \tilde{\mathbf{X}}^T\tilde{\mathbf{X}})^{-1}(\tilde{\mathbf{X}}^T\tilde{\mathbf{y}})$

Question 12

For what $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{y}}$ will the solution of this linear regression equal to

$$\mathbf{w}_{\text{reg}} = \text{argmin}_{\mathbf{w}} \frac{\lambda}{N} \|\mathbf{w}\|^2 + \frac{1}{N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 ?$$

- $\tilde{\mathbf{X}} = \mathbf{I}$, $\tilde{\mathbf{y}} = \mathbf{0}$
- $\tilde{\mathbf{X}} = \sqrt{\lambda}\mathbf{X}$, $\tilde{\mathbf{y}} = \mathbf{y}$
- $\tilde{\mathbf{X}} = \lambda\mathbf{I}$, $\tilde{\mathbf{y}} = \mathbf{1}$
- $\tilde{\mathbf{X}} = \sqrt{\lambda}\mathbf{I}$, $\tilde{\mathbf{y}} = \mathbf{0}$

Question 13

Experiment with Regularized Linear Regression and Validation

Consider regularized linear regression (also called ridge regression) for classification.

$$\mathbf{w}_{\text{reg}} = \text{argmin}_{\mathbf{w}} \frac{\lambda}{N} \|\mathbf{w}\|^2 + \frac{1}{N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2,$$

Run the algorithm on the following data set as $\mathcal{D}$

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw4%2Fhw4_train.dat

and the following set for evaulating E_{out}

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw4%2Fhw4_test.dat

Because the data sets are for classification, please consider only the $0/1$ error for all the problems below.
Let $\lambda = 10$, which of the followings is the corresponding E_{in} and E_{out}?

- $E_{in} = 0.035$, $E_{out} = 0.020$
- $E_{in} = 0.050$, $E_{out} = 0.045$
- $E_{in} = 0.015$, $E_{out} = 0.020$
- $E_{in} = 0.030$, $E_{out} = 0.015$

Question 14

Among $\log_{10}\lambda = \{2, 1, 0, -1, \ldots, -8, -9, -10\}$. What is the λ with the minimum E_{in}? Compute λ and its corresponding E_{in} and E_{out} then select the closest answer. Break the tie by selecting the largest λ.

- $\log_{10}\lambda = -4, E_{in} = 0.015, E_{out} = 0.020$
- $\log_{10}\lambda = -6, E_{in} = 0.030, E_{out} = 0.040$
- $\log_{10}\lambda = -10, E_{in} = 0.030, E_{out} = 0.040$
- $\log_{10}\lambda = -8, E_{in} = 0.015, E_{out} = 0.020$

Question 15

Among $\log_{10}\lambda = \{2, 1, 0, -1, \ldots, -8, -9, -10\}$. What is the λ with the minimum E_{out}? Compute λ and the corresponding E_{in} and E_{out} then select the closest answer. Break the tie by selecting the largest λ.

- $\log_{10}\lambda = -5, E_{in} = 0.015, E_{out} = 0.030$
- $\log_{10}\lambda = -9, E_{in} = 0.030, E_{out} = 0.030$
- $\log_{10}\lambda = -3, E_{in} = 0.015, E_{out} = 0.015$
- $\log_{10}\lambda = -7, E_{in} = 0.030, E_{out} = 0.015$

Question 16

Now split the given training examples in $\mathcal{D}$ to the first 120 examples for $\mathcal{D}_{train}$ and 80 for $\mathcal{D}_{val}$.

Ideally, you should randomly do the $120/80$ split. Because the given examples are already randomly permuted, however, we would use a fixed split for the purpose of this problem.

Run the algorithm on $\mathcal{D}_{train}$ to get g^-_λ, and validate g^-_λ with $\mathcal{D}_{val}$.
Among $\log_{10}\lambda = \{2, 1, 0, -1, \ldots, -8, -9, -10\}$. What is the λ with the minimum $E_{train}(g^-_\lambda)$? Compute λ and the corresponding $E_{train}(g^-_\lambda)$, $E_{val}(g^-_\lambda)$ and $E_{out}(g^-_\lambda)$ then select the closet answer. Break the tie by selecting the largest λ.

- $\log_{10}\lambda = -4, E_{train}(g_\lambda^-) = 0.000, E_{val}(g_\lambda^-) = 0.010, E_{out}(g_\lambda^-) = 0.035$
- $\log_{10}\lambda = -8, E_{train}(g_\lambda^-) = 0.000, E_{val}(g_\lambda^-) = 0.050, E_{out}(g_\lambda^-) = 0.025$
- $\log_{10}\lambda = -2, E_{train}(g_\lambda^-) = 0.010, E_{val}(g_\lambda^-) = 0.050, E_{out}(g_\lambda^-) = 0.035$
- $\log_{10}\lambda = -6, E_{train}(g_\lambda^-) = 0.010, E_{val}(g_\lambda^-) = 0.010, E_{out}(g_\lambda^-) = 0.025$

Question 17

Among $\log_{10}\lambda = \{2, 1, 0, -1, \ldots, -8, -9, -10\}$. What is the λ with the minimum $E_{val}(g_\lambda^-)$? Compute λ and the corresponding $E_{train}(g_\lambda^-)$, $E_{val}(g_\lambda^-)$ and $E_{out}(g_\lambda^-)$ then select the closet answer. Break the tie by selecting the largest λ.

- $\log_{10}\lambda = -6, E_{train}(g_\lambda^-) = 0.066, E_{val}(g_\lambda^-) = 0.038, E_{out}(g_\lambda^-) = 0.038$
- $\log_{10}\lambda = -9, E_{train}(g_\lambda^-) = 0.033, E_{val}(g_\lambda^-) = 0.028, E_{out}(g_\lambda^-) = 0.028$
- $\log_{10}\lambda = -3, E_{train}(g_\lambda^-) = 0.000, E_{val}(g_\lambda^-) = 0.028, E_{out}(g_\lambda^-) = 0.038$
- $\log_{10}\lambda = 0, E_{train}(g_\lambda^-) = 0.033, E_{val}(g_\lambda^-) = 0.038, E_{out}(g_\lambda^-) = 0.028$

Question 18

Run the algorithm with the optimal λ of the previous problem on the whole $\mathcal{D}$ to get g_λ. Compute $E_{in}(g_\lambda)$ and $E_{out}(g_\lambda)$ then select the closet answer.

- $E_{in}(g_\lambda) = 0.045, E_{out}(g_\lambda) = 0.030$
- $E_{in}(g_\lambda) = 0.025, E_{out}(g_\lambda) = 0.030$
- $E_{in}(g_\lambda) = 0.035, E_{out}(g_\lambda) = 0.020$
- $E_{in}(g_\lambda) = 0.055, E_{out}(g_\lambda) = 0.020$

Question 19

Now split the given training examples in $\mathcal{D}$ to five folds, the first 40 being fold 1, the next 40 being fold 2, and so on. Again, we take a fixed split because the given examples are already randomly permuted.

Among $\log_{10} \lambda = \{2, 1, 0, -1, \ldots, -8, -9, -10\}$. What is the λ with the minimum E_{cv}, where E_{cv} comes from the five folds defined above? Compute λ and the corresponding E_{cv} then select the closet answer. Break the tie by selecting the largest λ.

- $\log_{10} \lambda = -6, E_{cv} = 0.020$
- $\log_{10} \lambda = -4, E_{cv} = 0.030$
- $\log_{10} \lambda = -2, E_{cv} = 0.020$
- $\log_{10} \lambda = -8, E_{cv} = 0.030$

Question 20

Run the algorithm with the optimal λ of the previous problem on the whole $\mathcal{D}$ to get g_λ. Compute $E_{in}(g_\lambda)$ and $E_{out}(g_\lambda)$ then select the closet answer.

- $E_{in}(g_\lambda) = 0.035, E_{out}(g_\lambda) = 0.030$
- $E_{in}(g_\lambda) = 0.015, E_{out}(g_\lambda) = 0.020$
- $E_{in}(g_\lambda) = 0.005, E_{out}(g_\lambda) = 0.010$
- $E_{in}(g_\lambda) = 0.025, E_{out}(g_\lambda) = 0.020$

In accordance with the Coursera Honor Code, I (Sandipan Dey) certify that the answers here are my own work.

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